

Ahmed Bennasser

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 🌐 LinkedIn 🐙 GitHub 🌐 Portfolio

Computer Science and Applied Mathematics engineering student at ENSIMAG, focused on Machine Learning, Statistical Learning, and NLP, with a particular interest in LLMs and Generative AI. Experience in fine-tuning, Transformers, RAG, and model evaluation.

I am seeking a 5 to 6-month End-of-Study internship in AI/LLM/NLP/RAG starting January 2027.

EDUCATION

2024 – Present

Grenoble INP – Ensimag & Université Grenoble Alpes

- **Master 2 MSIAM** – MSc in Industrial and Applied Mathematics (2026–2027)
- **Engineering Degree** – Mathematical Modeling, Image and Simulation track.

2022 – 2024

Lycée Mohammed VI d'Excellence

Intensive two-year undergraduate program in Mathematics & Physics (CPGE)

EXPERIENCE

June 2026 – July 2026 Assistant Engineer Internship – LIG / TIMC, Grenoble

- Reimplementation of the *Hierarchical Feature Engineering* method leveraging taxonomic structure on sparse count data (microbiome data).
- Design and implementation of a statistical algorithm (**SFE**) for feature selection, evaluated against **HFE** and reference models across 5 experimental protocols: consistency, within-study, cross-study, noise stability, and high-dimensional regime ($p \gg n$).
- **Technologies:** Python, Scikit-learn, NumPy, Pandas, Statistical Analysis, Feature Engineering.

February 2026 – May 2026 Research Project – LIG (GETALP Team), Grenoble

- Analysis and evaluation of the French multimodal encoder *Pantagruel* (LIG) on NLP and NER tasks.
- Design of a text-to-pictogram generation pipeline: paired dataset creation, fine-tuning of diffusion models via **LoRA**, **Prompt Engineering** and **inference**.
- Development of a hybrid evaluation framework combining mathematical style and contrast metrics with semantic validation (*LLM-as-a-Judge*).
- **Technologies** : Python, TAL/NLP, NER, Transformers, PEFT, Quantization, JEPa, Word2Vec, LoRA, Stable Diffusion, Reward Models, LLM-as-a-Judge, Prompt Engineering. [GitHub](#)

PERSONAL PROJECTS

June 2026 – Present AgriGuard-RAG — RAG System for Agricultural Regulations

- Development of a RAG system for regulatory queries on plant protection products, with 5 277 structured and indexed documents.
- Design of a hybrid router combining heuristic rules and **TF-IDF + Logistic Regression** classification to dynamically route queries.
- Implementation of **hybrid retrieval** combining semantic embeddings (Sentence-Transformers), **BM25**, and metadata filtering with **Qdrant**.
- Planned extension toward an **agentic architecture** to handle complex queries through **reasoning workflows** and specialized tools such **LangChain/Prompt Engineering**.
- **Technologies:** LLM, Python, Scikit-learn, Qdrant, BGE, BM25, RAG, Information Retrieval, LangChain. [GitHub](#)

November 2025 Climate Sentiment Analysis

- Sentiment analysis applied to texts from Twitter related to climate issues.
- Comparative benchmark (RNN, LSTM, BERT): performance analysis (F1-score, precision, recall) illustrating BERT's advantage on large datasets and LSTM's robustness on limited data.
- Technologies: PyTorch, BERT, RNN, LSTM, Hugging Face Transformers, Pandas. [GitHub](#)

September 2025 Chest X-Ray Medical Diagnosis

- Classification of more than 20 000 medical images into four categories, with data augmentation and the use of CNNs as well as Transfer Learning methods based on ResNet18 and EfficientNet.
- Comparative evaluation of ResNet-18, EfficientNet, and a CNN developed from scratch, highlighting the performance gains provided by Transfer Learning. [GitHub](#)

SKILLS & LANGUAGES

Languages & Tools

Python • Java • C • C++ • SQL • R • Git • Linux • Jupyter • VS Code • LaTeX • VHDL • Assembly

AI & Machine Learning & Data Science

PyTorch • Scikit-learn • LLM • Transformers • Hugging Face • Qdrant • LangChain/LlamaIndex • BM25 • NumPy
• Pandas • Feature Engineering • Data Analysis • Prompt Engineering • Multimodal models

Arabic: Native | **French:** Fluent | **English:** Professional (scientific reading, technical writing)